
Using an MLST allelic profile to find a unique genotype identifier number

Introduction

For *Toxoplasma gondii*, the common multilocus genotyping approach uses a panel of markers such as **SAG1**, **SAG2**, **SAG3**, **BTUB**, **GRA6**, **c22-8**, **c29-2**, **L358**, **PK1**, and **Apico**. Historically, these markers have been typed using **PCR-RFLP**, in which allele types are assigned based on differences in restriction-enzyme digestion patterns.

These days, an alternative **in silico approach** called **multilocus sequence typing (MLST)** can be used when sequence data is available. Sequences from the marker loci are analyzed computationally to determine the allele at each locus directly and generate a multilocus genotype profile for each isolate.

This tutorial describes how to use an isolate's multilocus genotype profile to determine its corresponding unique genotype number.

Example

Your sample may have the following multilocus genotype profile

Sample01

SAG1	5' + 3' SAG2	Alt. SAG2	SAG3	BTUB	GRA6	C22-8	C29-2	L358	PK1	APICO
I	I	I	III	I	II	u-1	I	I	III	III

Step 1: Access the **genotypes table** at this link

https://beta.toxodb.org/toxo.beta/app/workspace/analyses/EDAUD_BNR_5Y9Zx5R0FA/new/variables/ENT_275a7000/VAR_84064fd6

The screenshot shows the ToxoDB Dataset Explorer interface. At the top, there's a search bar and navigation links. The main section is titled "Dataset Explorer" and "Table of unique PCR-RFLP genotypes". Below this, there's a "Records" box showing "354 of 354". The interface is divided into several tabs: "View Dataset Details", "Browse and Subset", "Visualize", "Download", "Record Notes", and "Learn". The "Browse and Subset" tab is active, showing a list of variables on the left and a table of records on the right. The variable "SAG1 genotype" is selected. The table shows the distribution of alleles for SAG1 genotype across 350 records (99% of 354 records have data for this variable).

Record	Subset of Records	All Records	Distribution	%
☐ I	246 (70%)	246 (70%)		(100%)
☐ II or III	47 (13%)	47 (13%)		(100%)
☐ u-1	56 (16%)	56 (16%)		(100%)
☐ u-2	1 (< 1%)	1 (< 1%)		(100%)

Notice the list of loci on the left. When you click on each locus name, you will see a list of alleles on the right.

Step 2: Go through the list and **for each locus, click the allele type that your sample corresponds to**. Repeat for all loci.

For instance-

- For **SAG1 genotype**, click "I"

This screenshot shows the "SAG1 genotype" variable selected in the "Browse and Subset" tab. The left sidebar lists various loci, with "SAG1 genotype" highlighted. The right panel shows the "1 row selected" table with the "I" allele selected. The table shows the distribution of alleles for SAG1 genotype across 350 records (99% of 354 records have data for this variable).

Record	Subset of Records	All Records	Distribution	%
☒ I	246 (70%)	246 (70%)		(100%)
☐ II or III	47 (13%)	47 (13%)		(100%)
☐ u-1	56 (16%)	56 (16%)		(100%)
☐ u-2	1 (< 1%)	1 (< 1%)		(100%)

- For the **5' + 3' SAG2 genotype**, click "I"

This screenshot shows the "5' + 3' SAG2 genotype" variable selected in the "Browse and Subset" tab. The left sidebar lists various loci, with "5' + 3' SAG2 genotype" highlighted. The right panel shows the "1 row selected" table with the "I" allele selected. The table shows the distribution of alleles for 5' + 3' SAG2 genotype across 350 records (99% of 354 records have data for this variable).

Record	Subset of Records	All Records	Distribution	%
☒ I	246 (70%)	246 (70%)		(100%)
☐ II	47 (13%)	47 (13%)		(100%)
☐ III	56 (16%)	56 (16%)		(100%)
☐ nd	1 (< 1%)	1 (< 1%)		(100%)

Step 3: Finally, you're at the **PCR-RFLP genotype** variable. Do three steps as indicated in the figure below. You're done!

Records
1 of 354

SAG1 genotype 5' + 3' SAG2 genotype Alt. SAG2 genotype SAG3 genotype BTUB genotype GRA6 genotype C22-8 genotype C29-2 genotype L358 genotype

View Dataset DetailsBrowse and SubsetVisualizeDownloadRecord NotesLearn

expand all | collapse all

Find a variable

Record

SAG1 genotype

5' + 3' SAG2 genotype

Alt. SAG2 genotype

SAG3 genotype

BTUB genotype

GRA6 genotype

C22-8 genotype

C29-2 genotype

L358 genotype

PK1 genotype

APICO genotype

PCR RFLP Genotype ID

(1) click on the PCR-RFLP Genotype ID variable

PCR RFLP Genotype ID

Original variable name: PCR_RFLP_Genotype

Unique identifier number assigned to each distinct multi-locus PCR-RFLP genotype in the dataset. A few numbers are not assigned.

Check items below to apply this filter

(2) click on the subset ordering button TWICE

PCR RFLP Genotype ID

Find items

Subset of Records

1 (100%)

1 (100%)

0 (0%)

0 (0%)

0 (0%)

0 (0%)

0 (0%)

0 (0%)

0 (0%)

0 (0%)

0 (0%)

(3) Note the PCR RFLP Genotype ID number for your multi-locus allele set

81

99

98

97

96

95

94

93

92